Calculating electron mobility for III–V alloys In($As_{1-x}Sb_x$), Ga($As_{1-x}Sb_x$) and ($In_{1-x}Ga_x$)($As_{1-x}Sb_x$).

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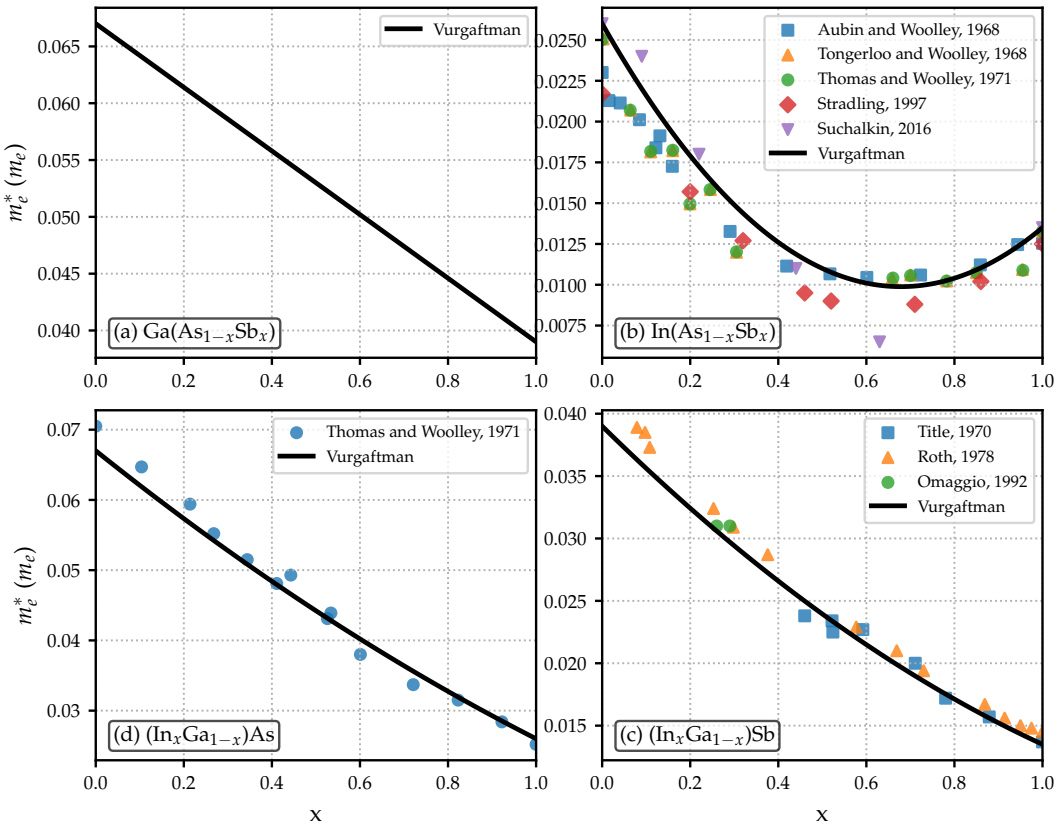


FIG. 1: The calculated electron mobility for In($As_{1-x}Sb_x$) (a), Ga($As_{1-x}Sb_x$) (b) and ($In_{1-x}Ga_x$)($As_{1-x}Sb_x$) (c) at 300 K, with the alloy scattering contribution shown as a dashed line.

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TABLE I: Fitted parameters and material properties. All experimental results are taken from the Ioffe institute. The F's are the factors associated with the Fröhlich interaction in

Material	x	y	U_A	U_B	m^*_{DFT}	m^*_{exp}	$\varepsilon_{\infty,\mathrm{DFT}}$	$\varepsilon_{\infty,\mathrm{exp}}$	$\omega_{\mathrm{TO,DFT}}$	$\omega_{\mathrm{TO,exp}}$	$\omega_{\mathrm{LO,DFT}}$	$\omega_{\mathrm{LO,exp}}$	$F^{\mathrm{Fr}}_{\mathrm{DFT}}$	$F^{\mathrm{Fr}}_{\mathrm{exp}}$	$F^{\mathrm{Fr}}_{\mathrm{DFT}}/F^{\mathrm{Fr}}_{\mathrm{exp}}$
GaAs	0	0	−3.73	5.84	0.073	0.067	8.81	10.89	7.52	8.02	8.89	8.55	0.29	0.09	3.04
$(In_{0.25}Ga_{0.75})As$	0.25	0	−2.69	4.66	0.055	0.055	9.41	–	7.15	–	8.31	–	0.23	0.09	2.46
$(In_{0.375}Ga_{0.625})As$	0.375	0	−2.31	4.39	0.050	0.050	9.55	–	7.15	–	8.24	–	0.21	0.09	2.31
$(In_{0.5}Ga_{0.5})As$	0.5	0	−1.80	4.00	0.044	0.044	9.87	–	6.50	–	7.60	–	0.21	0.09	2.25
$(In_{0.625}Ga_{0.375})As$	0.625	0	−1.26	3.72	0.039	0.039	10.10	–	6.69	–	7.69	–	0.19	0.09	2.04
$(In_{0.75}Ga_{0.25})As$	0.75	0	−0.36	3.23	0.034	0.035	10.57	–	6.66	–	7.58	–	0.16	0.09	1.81
InAs	1	0	−0.93	3.26	0.028	0.027	10.74	12.30	6.14	6.44	7.10	7.01	0.17	0.09	1.86
$In(As_{0.75}Sb_{0.25})$	1	0.25	−1.87	3.15	0.016	0.016	11.90	–	6.09	–	6.80	–	0.11	0.07	1.53
$In(As_{0.625}Sb_{0.375})$	1	0.375	−1.89	3.08	0.013	0.013	12.43	–	5.48	–	6.17	–	0.10	0.07	1.54
$In(As_{0.5625}Sb_{0.4375})$	1	0.4375	−1.36	2.83	0.011	0.012	12.98	–	5.68	–	6.29	–	0.09	0.07	1.38
$In(As_{0.5}Sb_{0.5})$	1	0.5	−1.97	3.07	0.011	0.011	12.88	–	5.28	–	5.92	–	0.09	0.06	1.52
$In(As_{0.375}Sb_{0.625})$	1	0.625	−1.47	2.91	0.010	0.010	13.51	–	5.52	–	6.07	–	0.08	0.06	1.37
$In(As_{0.25}Sb_{0.75})$	1	0.75	−2.78	3.48	0.010	0.010	13.10	–	5.00	–	5.61	–	0.09	0.05	1.68
InSb	1	1	−1.10	3.16	0.018	0.014	13.30	15.70	5.17	5.54	5.67	5.90	0.07	0.04	1.64